

SAURABH SINGH



A Parallel Universe

A complex mathematical puzzle that hovers on the brink of brilliance



A CERTAIN AMBIGUITY: A MATHEMATICAL NOVEL
by GAUTAM SURI, HARTOSH SINGH BAL
Viking
Price: Rs 450; Pages: 282

■ By Geeta Doctor

Is mathematics the new mantra? For those who dream of infinity and drool over the decimal point, Gautam Suri and Hartosh Singh Bal have created a complex mathematical puzzle that hovers on the brink of brilliance, without quite getting there. Perhaps, that is the whole point about infinity. It's always just beyond the reach of the human mind.

Maybe, as Douglas Hofstadter explained with brilliant felicity in the late 1970s when he wrote *Gödel, Escher, Bach: An Eternal Golden Braid*, it is this very paradox—the limitless limits of the human imagination—that makes us human. Hofstadter is a Stanford alumnus, like the narrator of *A Certain Ambiguity*, who comes through as a bit of a nerd named Ravi. A world of perceptions separates the real-life Hofstadter from the imaginary Ravi, though they share the Stanford whiz kid's prerogative—a tendency to show off. While Hofstadter dazzles, plucking off puzzles, Ravi plods. Though Suri and Bal explain that they want to bring out the beauty that lies in mathemat-

ics, their theories crisscross in a distinctly non-Euclidean manner.

One set of narratives focuses on the lectures that Ravi and friends attend. Their teacher is a charismatic professor, Nico, who intones, "I can only say that God is a mathematician." He distills the wisdom of the ages, allowing both the students and the reader to work out various mathematical equations. There are walk-on parts for famous theoreticians and thinkers of the past. Our own Bhaskara sounds wooden, as do most of the other great minds, who are supposedly sharing their innermost thoughts in the form of diary entries.

The second parallel plot takes the story back to Ravi's grandfather, Vijay Sahni, a mathematician so brilliant he was invited to the US around 1919. Sahni created such uproar among the Bible-thumping citizens of New Jersey that they put him in jail for blasphemy. Ravi follows his *bauji's* ordeal via old transcripts of the protracted battle, fought along Euclidean axioms that he demonstrates to his adversary, a crusty old judge named Taylor. Their battle seems to be between holism and reductionism, but just as Sahni seems to be winning, Taylor manages to trump him. The conclusion is no different from the one arrived at by Hofstadter: the brain is a far more complex place than even it can imagine. ■

Public Space

An engaging chronicle of how satellite missions impact India



TOUCHING LIVES: THE LITTLE KNOWN TRIUMPHS OF THE INDIAN SPACE PROGRAMME
by S.K. DAS
Penguin
Price: Rs 250; Pages: 251

■ By Raj Chengappa

The Indian Space Research Organisation (ISRO) has been in the public eye for its ability to indigenously build satellites and then launch them with spectacular success. Few know that the country's space mission impacts the lives of most Indians in ways that are more obvious than satellite television or telephony. Applying sophisticated technology to enhance the country's development is an area where India is acknowledged as a leader by the space powers of the world. One reason for the lack of appreciation in the country is that it tends to be all about, well, rocket science—a dense subject which people find difficult to comprehend. *Touching Lives*, a chronicle of ISRO's success in space applications, fills that void by taking the reader on an educative tour across the country and seeing it through the eyes of the people who benefited directly from them.

Written by S.K. Das, a retired IAS officer who served in the Department of Space, the book is shorn of jargon and is a simple but engaging narrative of how far-flung villages gain from educational programmes telecast using ISRO satellites and how its remote sensing programme allows fishermen to boost their catch, helps villagers dig bore wells with far greater success and empowers panchayats with data dossiers. Das energetically questions villagers and makes pertinent observations, laced with wry humour, which makes the book an enjoyable read. The only failing is that the author takes an uncritical look at the programme. There is a need for an overall assessment and a look into what more can be done to maximise the programme's benefits. That would have enhanced the book's value. ■